

Ucaneo & CO-reactive

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Ucaneo Biotech GmbH & Co-reactive GmbH

Company or organization location (we welcome applicants from anywhere in the world)

Berlin & Duesseldorf, Germany

Name(s) of primary point(s) of contact for this application

Florian Tiller & Dr. Andreas Bremen

Brief company or organization description <20 words

Ucaneo and Co-reactive combine electrochemical DAC and engineered mineralization to turn CO₂ into affordable supplementary cementitious materials.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

Project summary

The joint Ucaneo x Co-reactive project couples a novel flexible load electrochemical Direct Air Capture (DAC) technology with an innovative continuous, highly scalable CO₂ mineralization route. Ucaneo removes CO₂ directly from ambient air using a novel electrodialysis pH-swing process with high energy (<1000- 1500 kWh/CO₂t) and cost efficiency. The captured CO₂ is then transported by truck to Co-reactive's facility, where it is permanently (>1,000 years) sequestered as carbonate rock to produce supplementary cementitious materials (SCMs), delivering durable decarbonization for the building sector.

Together, they target both prioritized pre-purchase innovation areas by Frontier: "Efficient and flexible energy use" as well as "Accelerating in-situ and ex-situ mineralization for CO₂ storage."

Ucaneo's patent pending DAC technology uses an **electrodialysis pH swing** combined with bio-inspired materials and tailored component designs to achieve efficient, flexible-load, low-temperature CO₂ removal. The process comprises of **three steps** (see Figure 1):

1. **Air contact and absorption:** Ambient air is contacted with a custom liquid air contactor. CO₂ dissolves into Ucaneo's aqueous (bio)catalytic solvent and is stored predominantly as pH-dependent bicarbonate ions.
2. **pH swing separation:** Two independent solvent streams are used to maximize the pH swing: an alkaline stream optimized for absorption and an acidic stream optimized for desorption. The streams are kept distinct by a custom electrodialysis (ED) stack containing bipolar membranes, which maintains stable pH levels and enables efficient transfer.
3. **CO₂ release and conditioning:** Acidification and vacuum degassing release high-purity CO₂ in a dedicated degassing unit. The CO₂ is then conditioned for transport to Co-reactive's mineralization site.

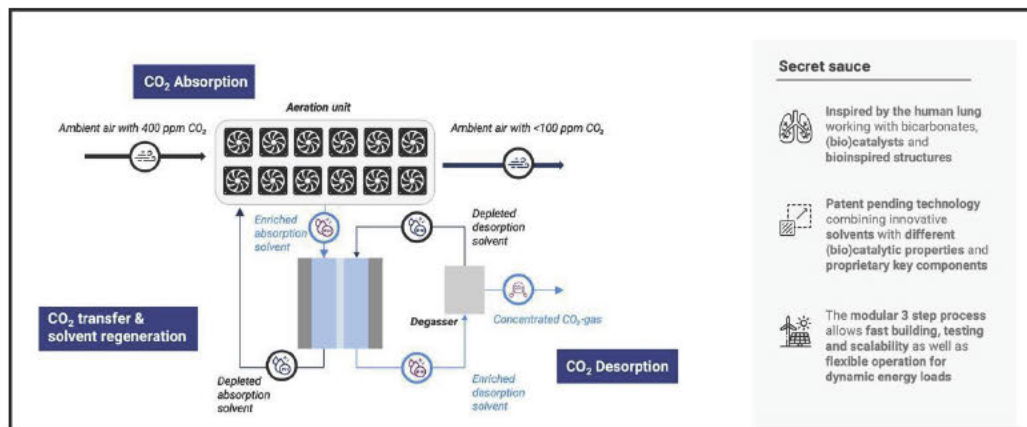


Figure 1: Ucaneo's electrochemical low-cost and energy flexible DAC Process

Advantages of Ucaneo's patent-pending novel electrochemical technology include:

1. **Cost efficiency:** Very low levelized costs of carbon removal of around \$80-90 / tCO₂ at NOAK level (100 ktCO₂/y). Supported by a third-party feasibility study by [TTP Cambridge](#), Ucaneo is able to reach costs of <\$300 / tCO₂ already at >1000 tCO₂/y level by 2028/2029.
2. **Energy efficiency:** Overall already 1,000-1,500 kWh / tCO₂ validated energy efficiency on an industrial pilot level (50 tCO₂/y) with NOAK targeted energy efficiency of 800-1,000 kWh/tCO₂. Energy efficiency of <1,000 kWh/tCO₂ for regeneration can already be indicated by lab data.
3. **Flexible load operations & renewable integration:** Ucaneo's electrochemical system for desorption can be quickly ramped up and down reliably plus allows storage of CO₂ / indirectly chemical energy in liquid solvents, while absorption can operate continuously on low energy demand. This enables Ucaneo to optimize operation of its systems flexibly depending on electricity availability and energy prices. That flexibility creates opportunities for energy-price arbitrage and makes the technology especially attractive in regions with high grid prices or remote renewable generation — reducing effective OPEX and improving project economics.
4. **Rapid scalability:** Ucaneo leverages mature engineering knowledge (water desalination, acid/base recovery), solely widely available materials and existing supply chains, augmenting them with proprietary (bio)catalytic solvent compositions, liquid air contactor designs as well as process and cell-stack designs. Thanks to their proprietary stack and process design, Ucaneo can achieve those competitive energy efficiencies using even the lowest-cost membranes available on the market. Overall, this levered approach allows accelerated commercialization, allows deployment at scale faster and at much lower cost from small to gigaton scale.
5. **Carbon efficient operations:** As the process operates in a closed loop system solely powered by electric energy, it can be powered directly by variable renewables, such as solar or wind. This enables low operational emissions and contributes to a high removal efficiency, allowing optimal carbon-negativity.
6. **Modular architecture:** The design is inherently modular, enabling incremental, low-risk scale-up. Modules can be deployed and validated in parallel, matched to locally available energy sources, and integrated with diverse downstream CO₂ handling options — reducing capital risk and shortening the path from pilot to commercial array. This also allows to tailor the CAPEX / OPEX balance depending on the project's main objectives, e.g. using higher cost membranes to reduce energy need.
7. **Nontoxicity & environmental standards:** Ucaneo's solvents are non-toxic and use widely available additives. Developing the solution in Germany, strict care is paid to environmental protection.

Ucaneo's industrial pilot (up 50 tCO₂/y, TRL 6) demonstrated that the technology is already among the world's most energy- and cost-efficient DAC solutions. With continued development, Ucaneo continues to improve its process long-term to energy intensities of 800-1,000 kWh/tCO₂ and a cost floor near \$80-90 / tCO₂ at NOAK level (100 ktCO₂/y).

Co-reactive's process permanently transforms captured CO₂ into stable mineral carbonates to

produce high-performance CO-SCMs using a patent-pending continuous tubular CO-reactor. The well-known storage potential of CO₂ as mineral carbonates (>1,000 years permanence) is lifted to a commercially viable process using the Co-reactive technology. This integrated approach not only enables cost-effective CO₂ storage, but also delivers the scalable production of urgently needed, carbon-efficient construction materials.

The **Co-reactive process** employs a wet, direct mineralization process that reacts captured gaseous CO₂ with abundant magnesium/calcium silicate feedstocks, e.g., olivine or metallurgical slags, in an aqueous slurry according to the following stoichiometry:



This process yields solid carbonates (MgCO₃ - magnesite / CaCO₃ - calcite) that lock CO₂ in mineral form, and amorphous silica (SiO₂, up to ≈25 wt%), a so-called pozzolan that enhances hydration (hardening) of cementitious systems and mechanically binding carbonates within the concrete matrix. The process consists of **five steps** (see Figure 2):

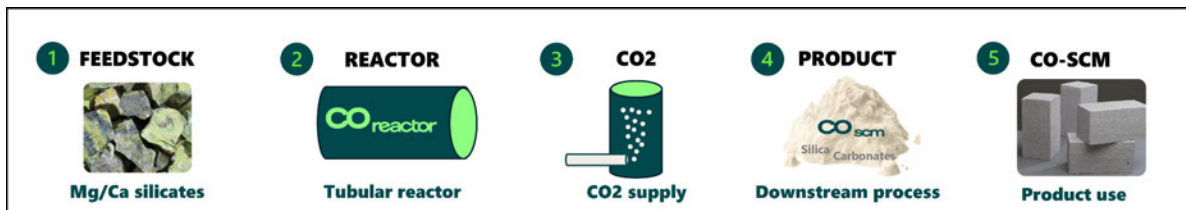


Figure 2: Co-reactive's Mineralization Utilization Process Steps

- 1. Feedstock:** Feedstock materials (e.g., primary minerals like olivine or metallurgical slags like EAF, BOF, and ladle slags) are milled to a fine powder and suspended into an aqueous slurry containing inorganic additives. Strategic partnerships secure local feedstock streams and reduce transport and preprocessing footprint.
- 2. Continuous tubular CO-reactor:** The slurry is pumped through a high-temperature, high-pressure tubular reactor composed of upward and downward pipes designed to maximize heat transfer, prevent sedimentation by aligning flow with gravity, and optimize flow dynamics—achieving 98% CO₂ mineralization efficiency without internal moving parts.
- 3. CO₂ delivery:** Captured CO₂ delivered by Ucaneo is injected through nozzles at the bottom of the pipes, thus generating a high surface area between aqueous and gas phase to enhance mass transfer and mixing of phases. CO₂ is injected in an over-stoichiometric manner - unreacted CO₂ rises to the top of the pipes and is recycled through a recycling pump for maximum conversion.
- 4. Solid-liquid separation and drying:** Carbonated solids are dewatered and dried in state-of-the-art equipment (e.g., filter press & flash dryer) using recovered reaction heat to obtain a dry, fine powder that meets SCM industry standards.
- 5. Supply to cement/construction:** CO-SCMs are supplied to high-value concrete and construction chemicals applications, such as dry mortars and geopolymers (in the short-term), and as drop-in replacements for fly ash / ground granulated blast furnace slag (GBFS) in

ready-mix concrete, precast products, and cement formulations (in the long-term).

Scalability constraints inherent to autoclave batch systems are the principal barrier to delivering mineralized SCMs at industrial scale. While some competitors produce aggregates, employ carbon-curing, or rely on autoclave carbonation, these approaches are limited by batch downtime, low space-time yield, dead volumes and incomplete carbonation—driving higher CAPEX/OPEX and producing materials with limited pozzolanic reactivity unsuited for broad cement substitution.

Advantages of Co-reactive's patent-pending novel mineralization approach includes:

- 1. High-throughput, continuous tubular reactor:** High-throughput, continuous reactor design. Co-reactive's continuous tubular CO-reactor runs 24/7 with a 2.5x higher space-time yield compared to autoclave benchmarks by eliminating batch downtime and dead zones. Prototype and modelling show near full carbonation of feedstock materials while having a CO₂ efficiency of up to 98%. The design enables modular scale-up from Co-reactive's demo plant to industrial scale.
- 2. Efficiency with validated product performance.** The exothermal reaction heat is recovered to supply thermal energy for drying, which is required for material application in cement & concrete. Produced CO-SCMs deliver high pozzolanic reactivity (approx. 25% amorphous SiO₂), thus exceeding strength activity benchmarks (100% strength activity index at 25% replacement of CEM I), and exhibit a strong carbon removal (-330 kg CO₂/t CO-SCM).
- 3. Feedstock flexibility:** Process validated on olivine and metallurgical slags (15 feedstock sources validated) with demonstrated pozzolanic reactivity of produced CO-SCMs; strategic feedstock partnerships and a low-maintenance, no-moving-parts reactor design reduce preprocessing and scale-up risk, supporting rapid industrial deployment.

The final CO-SCM products are thermodynamically more stable than gaseous CO₂ and remain intact under atmospheric conditions for timescales well beyond 1,000 years, with natural analogues demonstrating persistence over millions of years. Unlike CO₂-derived fuels, polymers, or chemicals that can re-release carbon during use or degradation, mineral carbonates do not revert. When applied in cement and concrete, the carbonates are further embedded within a solid matrix, enhancing durability and preventing leaching or breakdown throughout the product's life cycle and beyond demolition or recycling. In summary, the storage of carbonates incorporated into construction materials at atmospheric conditions does not require costly monitoring over long timescales.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

Project objectives

Ucaneo and Co-reactive are **jointly developing one of Europe's largest fully integrated electrochemical DAC and continuous mineralization chains.**

Germany is currently blocked by deploying DAC due to high electricity grid prices and unclarity on geological storage perspective. Ucaneo's and Co-reactive's joint initiative demonstrates how climate impact and economic value can be achieved in tandem in Europe: delivering permanent CO₂ removal, reducing cement-sector emissions, and strengthening domestic SCM supply chains. Further, both companies provide critical innovations enabling the deployability of DAC in Germany. Ucaneo's flexible load operations & renewable integration further optimize the high electricity costs in Germany and can help Germany to balance its grid as Germany is facing one of the highest negative electricity prices in its history. Co-reactive's novel mineralization technology for CO₂ storage and its simultaneous utilization to produce supplementary cementitious materials (SCMs) represents a critical innovation. This is especially relevant given that large-scale CO₂ transport and storage infrastructure is still in its early stages, and subsurface storage faces significant regulatory challenges—particularly in countries like Germany. By integrating long-term carbon storage with scalable production of cementitious materials, Co-reactive offers a dual solution that is essential for decarbonizing the “hard-to-abate” construction materials sector.

The proposed joint project is uniquely derisked already through existing deployments and proven efficiencies on scale as well as the possibility to deliver tons through the combination of several deployments. Both companies' scale-up plans are aligning very well with the near-term goal to validate this project and de-risk the path to industrial deployment.

- Ucaneo is constructing its **technical FOAK plant (TRL 7)** in Berlin, commissioning in **Q2 2026**. It already started construction in August 2025. With a capture capacity of **~150-365 tCO₂/y**, it will be one of Germany's largest DAC facilities and will demonstrate fully integrated operation. It goes beyond just a containerized pilot, but is truly a small industrial plant, whose design and partnerships can easily be scaled up. This builds on Ucaneo's **industrial pilot (TRL 6)** commissioned in **Q2 2025** at **~50 tCO₂/y**, which has already proven robustness and efficiency at final scale. Furthermore, Ucaneo is in the preparation phase for its **commercial FOAK (TRL 8/9)** in Germany, a scaled and further optimized version of the technical FOAK with better scaling effects which sustains as a commercial plant with industrial lifetime. It is scheduled to begin construction in 2027 with commissioning planned in **Q4 2027/Q1 2028**. With a capture capacity of **~1,000-1,500 tCO₂/y**, it will be one of Europe's largest DAC plants and will support commercial optimization while demonstrating the full value chain. For the NOAK the FOAK can be easily duplicated to the desired scale.
- Co-reactive is constructing its **demonstration plant (TRL 7)** in Düsseldorf, commissioning in **Q1 2026**. It will operate continuously utilizing **~710 tCO₂/y** to produce **~2,000 t of CO-SCM/y**, serving as an automated, 24/7 reference plant for customers and partners from concrete pavers to specialty chemicals allowing future scaling to a NOAK level as required. The capacity is planned to be increased with their **commercial FOAK** to **~3,000 tCO₂/y** to produce **~10,000 t of CO-SCM/y (3,000 t CO₂/y) by Q3 2027**. Co-reactive has already successfully developed and commissioned a **3 tCO₂/y** prototype to produce **10 t of CO-SCM/y in a semi-continuous pilot reactor (TRL 5)**, producing high-value CO-SCMs that exceed industry standards and have been tested with customers from concrete pavers and specialty chemicals to cement industry players looking for large-scale decarbonization

options.

Ucaneo, based in Berlin, builds its pilot and technical FOAK on their own premises in Berlin on leased land of more than 1000m² based in an existing industrial complex. The commercial FOAK is planned to be built at an industrial park within a few hours drive from Berlin (location not publicly announced yet). Co-reactive, based in Düsseldorf, builds their demonstration plant in their technical center in Erkrath (leased site). Its commercial FOAK is planned to be built near Düsseldorf on the premises of a ferroalloy manufacturer, providing direct access to locally available metallurgical slags. Co-reactive's strategic location in North Rhine-Westphalia offers a unique advantage by combining access to regional feedstocks (such as olivine and metallurgical slags), strong partnerships with established cement and concrete producers, efficient SCM distribution logistics, and proximity to a dense network of heavy industry plant engineering firms. Both Ucaeno's and Co-reactive's initial sites are intentionally designed at the sub-ktCO₂/y scale to validate continuous operation, automation, and product performance before expanding to multi-ktCO₂/y facilities. A potential transportation partner for the proposed project has already been identified. At commercial scale, Ucaneo and Co-reactive aim for co-location of capture and mineralization at a single site to eliminate CO₂ transport costs and further optimize system economics.

Current cost breakdown

The proposed project cost is \$1275 / tCO₂ including removal costs (\$919 / tCO₂), mineralization (\$276 / tCO₂) and transport (80\$ / tCO₂) in Germany. Capital costs are with up to 30 % one of main drivers of the cost breakdown, fixed operating costs especially labor costs up to 40% as well as energy costs with up to 15 %. To reach levelized costs of <\$100 / tCO₂ capital costs must be decreased by a factor of 4 (400%) which comes naturally with unlocking new type of financing, electrical energy prices must decrease from \$100 / MWh to \$40 / MWh as well the plant size increased to a 100 kt/y plant in order to reduce CAPEX cost due to economies of scale and the cost of labor per ton of CO₂.

MRV framework

The following MRV framework will be implemented:

1. **Mass balance and traceability:** (1) Calibrated flow meters measure CO₂ input, recycle streams, and off-gas at the outlet of Ucaneo's process, transport and inlet of Co-reactive, (2) time-stamped, continuous logging ensures full auditability across the process, and (3) enables end-to-end traceability from capture to permanent storage.
2. **CO₂ quality control:** (1) Continuous monitoring of purity and moisture content of the outlet of Ucaneo's process, transport and inlet of Co-reactive, (2) direct sensor integration for automated quality assurance, and (3) real-time anomaly detection with alerts to minimize off-spec risks.
3. **Carbon uptake and product analytics:** (1) Regular CO-SCM sampling analyzed via LOI, TC/TOC, TGA, and XRD, (2) stoichiometric cross-checks against feedstock Mg/Ca content confirm conversion, to ensure precise quantification of sequestered carbon in solid carbonate form by Co-reactive.

4. **Permanence verification:** (1) Mineralogical assays (XRD, SEM) confirm crystalline carbonate phases, (2) leachability and durability tests demonstrate long-term stability, and (3) concrete integration studies (e.g., Strength Activity Index, service-life durability) validate permanence in applied materials.
5. **Independent verification and LCA:** (1) ISO-aligned QA/QC with periodic third-party audits and certification, (2) accredited carbon removal certificates issued based on verified uptake, and (3) cradle-to-gate LCA incorporates avoided clinker emissions and prevents double counting.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

I. General risks

1. **Regulatory and market:** Slow bureaucratic approval processes in Germany, very high work safety and environmental standards in Germany, certification delays, legislative uncertainties around CO₂ transport and storage, evolving CDR legislation. | Mitigation: Early engagement with regulators especially around BImSchG, working with (public certified) expert planners and EPC companies already at early stage, working with standard bodies (e.g., EN 197/206, REACH/CE) with staged certification strategies, community support to ensure political support, and piloting in low-barrier product applications. Monitoring EU and German policy changes ensures proactive adaptation as well as being founding members of DVNE.
2. **Financial and market development:** Limited early market demand for CO-SCMs or investment shortfall could constrain scale-up. | Mitigation: Target high-margin niche applications (prefab, specialty mortars, geopolymers) for early revenue, secure offtake agreements with cement and concrete partners, leverage committed investors, staged financing, and blended finance to de-risk exposure. Already secured >€7 million in grants for this project.
3. **Environmental and health:** Co-reactive's feedstocks may contain heavy metals or impurities. | Mitigation: Prioritize non-problematic materials, pre-treat or blend marginal feedstocks, implement hydrometallurgical options, conduct leach testing, and comply with REACH and performance-based safety metrics.

II. Ucaneo (DAC) specific risks

1. **Technical and operational for FOAK project:** Delays or unavailability of critical components, or failure to meet technical specifications, could impact plant deployment. | Mitigation: Collaborate with experienced EPC partners, expert planners and component suppliers. Main supplier quotes are already available for the technical FOAK and construction of technical FOAK has already started with thorough expert planners and EPC work. Main design is based on very successful industrial pilot. Commercial FOAK will mainly be an upscale of the technical FOAK with minimal changes and leveraging real-world feedback loops.
2. **Process robustness and lifetime of membranes:** a) Continuous operation at scale over several months is untested beyond pilots. | Mitigation: Industrial pilot data informs FOAK

design. Design has been validated and built for the technical FOAK with an EPC partner together. Clear operational and maintenance guidelines that reduce variability have been created by very experienced O&G and process engineering team members; risk assessed as moderate. b) Lifetime of membranes is untested on industrial scale over several years | Mitigation: Only membranes are used which have been deployed in industry over several years providing good operational experience. Specifically, for Ucaneo’s process the membranes have gone through an accelerated stress testing, developed in collaboration with a university, proving they have the same lifetimes as seen in industry and fulfill the lifetime expectations according to the suppliers. c) Aeration capture rate and water loss is not fully validated on industrial scale across different seasons | Mitigation: Close engagement with supplier to do extended FATS with enough time to adjust design regarding channeling or water loss observations. Highlighting bespoke design brings risks but also potential to deliver performance goals for FOAK.

3. **Energy costs and economics:** High regional electricity costs could impact FOAK economics. | Mitigation: Energy-efficient, low-temperature electrochemical process reduces energy demand significantly. Unit economics have been validated before scale-up on pilot level already to reach proposed cost targets. Further, flexible load operations enable leveraging energy arbitrage especially in combination with smart energy supply. First partners for smart energy supply have been engaged.

III. Co-reactive (Mineralization) specific risks

1. **Technical process:** (a) Caking, scaling, solid precipitation. | Mitigation: Vertical flow reactor design, optimized slurry rheology, side CO₂ injection, scheduled cleaning, and spare modules for hot-swap replacements; (b) Component wear and limited lifetime | Mitigation: Robust materials, supplier qualification, accelerated lifetime testing, preventative maintenance, and redundancy; (c) Carbon balance underperformance | Mitigation: Integrated heat recovery, energy optimization, low-carbon feedstocks, and continuous LCA updates.
2. **Project execution and FOAK scale-up:** Scaling reactor geometry and balance-of-plant presents mechanical and control challenges. | Mitigation: Phased modular replication, digital twins for process control, iterative commissioning, and pre/post-processing validation.
3. **Ecosystem and market integration:** Slow standardization, incumbent resistance, and small initial plant economics. | Mitigation: Co-pilots with major industry partners, staged market entry, local feedstock sourcing, and leveraging avoided clinker costs to build economic rationale.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	392 tCO ₂
Delivery window	December 2028

<i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$2601 per tCO ₂
Levelized price (\$/ton CO ₂) ³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$1275 per tCO ₂

³ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).